BEDANTA BANDYOPADHYAY

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SUMMARY

I'm a final-year Electronics & Communication student who builds software end to end, from the interface down to the servers it runs on. My work spans full-stack, AI and IoT, and I like taking a project all the way from an idea to something people actually use. I'm also the solo founder and developer of HonorRoll, so I'm used to owning every part of a product myself.

EDUCATION

INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA — **Expected 2027**
B.Tech, Electronics & Communication Engineering · CGPA 8.56 (through 6th semester)

RAM MOHAN MISSION HIGH SCHOOL — **2019 – 2023**
ISC (Class XII) 73.2% · ICSE (Class X) 94.5%

SKILLS

FRONTEND	React, Vite, Tailwind CSS, HTML, CSS
BACKEND	Python, Java, Node.js, Express, FastAPI, Flask, Docker
DATABASE	PostgreSQL, SQL, Redis, Firebase
AI / ML	TensorFlow, PyTorch, scikit-learn, OpenCV, MediaPipe, LangChain, Local Inference (vLLM)
HARDWARE / IOT	C, Arduino, Raspberry Pi, ESP32
TOOLS	Git, GitHub Actions, Linux
FUNDAMENTALS	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)

EXPERIENCE

RESEARCH INTERN — IEDC, IEM Newtown — **Jun 2025 – Jul 2025**
Advanced & Cleaner Materials in Solar Technology

- Optimized the performance of tri-layer, lead-free, carbon-based perovskite solar cells using machine-learning models, and authored the full technical report.

PROJECTS

HONORROLL — Multi-Tenant LMS, Coding Judge & Proctored Exam Platform — **Jul 2026 – Present**
pandadsgn.github.io/HonorRoll

- Shipped a multi-tenant SaaS used by ~300 students across 4 roles (student, teacher, admin, superadmin), backed by 200+ REST endpoints and 50+ PostgreSQL tables, by building a Node.js/Express API with a React web app and a React Native/Expo mobile app.
- Ran untrusted student code in 10 languages on a host with no container privileges, capped per run (5 s CPU / 64 MB for Python and C), by replacing a Docker-per-run sandbox with unprivileged-user process isolation and ulimits.
- Made the backend horizontally scalable (nginx load-balancing across replicas, Redis-shared rate limits and cache) by containerizing the stack with Docker Compose, and gated every push with 129 integration tests against real PostgreSQL in GitHub Actions CI.
- Detected 4 classes of exam violation (face, gaze, phone, speech) in real time by running MediaPipe vision and audio models on-device in the browser with exam lockdown.

`React / React Native/Expo / Node.js / Express / PostgreSQL / Redis / Docker / S3 / MediaPipe / Razorpay / GitHub Actions`

CODEHAUNT — Self-Hosted, AI-Tutored Coding Contest Platform — **Nov 2026**
github.com/PandaDsgn/codehaunt-main

- Restored a live CSI UEMK contest mid-event after the primary server crashed by migrating the Docker Compose app tier (Next.js web, judge worker, PostgreSQL, code sandbox) to an NVIDIA DGX Spark, as owner of its server infrastructure.
- Served 50 concurrent AI-tutor sessions at ~1 s p95 time-to-first-token, down from up to 22 s in early load tests, by fronting 4 vLLM model servers on a k3s GPU cluster with a least-busy routing gateway that merges streamed tokens every 100 ms over a Tailscale link.
- Exposed the platform with zero inbound ports and a measured 33 s rollback by running a 2-connector Cloudflare Tunnel with WAF rules, tagged releases with expand-then-contract migrations, and pg_dump backups every 5 minutes to two hosts.

`Next.js / TypeScript / Node.js / PostgreSQL / Prisma / Docker Compose / vLLM / k3s / Cloudflare Tunnel + WAF / Tailscale / Safe Exam Browser / Playwright`

IDLESERVICEKILLER — Low-Footprint Windows Service Manager **Oct 2026**
github.com/PandaDsgn/IdleServiceKiller

- Freed CPU and RAM on Windows machines with a background service that itself holds ~1 MB of RAM, by writing a single-threaded Rust binary with one dependency that trims its own working set after every scan.
- Avoided breaking critical software by stopping only third-party services idle for 30 minutes (≤50 ms CPU and ≤64 KB I/O per 120 s scan) and skipping built-in deny lists (security, VPN, drivers, Microsoft-published), with dry-run as the default.

`Rust / Win32 API / PowerShell`

INFRASTRUCTURE-ASSASSIN — AI-Driven Cloud Resource Governance Platform **Jul 2026**
pandadsgn.github.io/Infrastructure-Assassin-Production

- Kept the AI analysis assistant answering through provider outages and quota limits with a 4-tier fallback chain (Gemini 2.5 Flash → Groq Llama 3.1 → DeepSeek → local rule-based engine) on a Python/FastAPI backend.
- Pushed resource and approval changes to every open dashboard without polling by fanning PostgreSQL LISTEN/NOTIFY events out over Server-Sent Events, with Redis holding per-user chat state.
- Enforced a 2-role approval workflow (Junior-Developer requests queue for IT-Director sign-off) across 15 API endpoints using Firebase ID-token auth and Firestore role lookups.

`Python / FastAPI / PostgreSQL / Redis / Firebase / React / Vite / Render / GitHub Pages`

OCCUNOVA — Deep Learning Diagnostic System for Glaucoma Detection **Jun 2026**
pandadsgn.github.io/OccuNova

- Reached 97.67% recall on glaucoma screening from retinal fundus scans by cascading two ResNet50 classifiers: a high-sensitivity triage model, then a specialist model trained with hard-negative mining.
- Eliminated manual region-of-interest cropping by training a U-Net that segments the optic disc and auto-crops a 224×224 input, served with both classifiers through a Dockerized FastAPI/TensorFlow inference API and a React frontend.

`Python / TensorFlow / OpenCV / FastAPI / Docker / React`

AGRO-GUARD — IoT Crop & Grain-Silo Monitoring System **Feb 2026**
pandadsgn.github.io/Agro-Guard

- Classified silo storage conditions as safe or spoilage-risk at 92.5% hold-out accuracy by training a 100-tree Random Forest on ~11,000 labelled readings (temperature, humidity, VPD, crop type) and serving it through a Flask API on Render.
- Streamed live telemetry from an Arduino Uno (DHT11, HC-SR04, LDR) to the cloud every ~2 s, with auto-reconnect and an 8 s stale-data cutoff, by writing a Python serial agent that pushes API-key-authenticated readings over HTTPS.
- Flagged sensor anomalies beyond 3σ and forecast time-to-empty for the silo by implementing z-score detection and linear regression in the React dashboard.

`Arduino / Python / Flask / scikit-learn / React / Node.js / Render`

LEADERSHIP

VICE PRESIDENT — CSI UEMK Student Chapter (Computer Society of India) **Jan 2026 – Present**

- Promoted to Vice President after serving as Manager (Aug 2025 – Jan 2026) and Graphics Lead (Mar 2025 – Jan 2026).
- Organised multiple chapter events and competitions as Vice President.

TECH / MEDIA / GRAPHICS LEAD — Iris, UEMK Photography Society **Sep 2024 – Aug 2025**

PUBLICATION

AgroGuard: A Novel IoT Enabled Framework for Smart Crop Preservation in Grain Silos
Bentham Science · Status: ongoing

CERTIFICATIONS

Cloud Computing — NPTEL, IIT Kharagpur **Jun 2025**
Business Intelligence & Analytics — NPTEL, IIT Madras **Jun 2025**
Joy of Computing with Python — NPTEL, IIT Ropar **Nov 2024**
Introduction to IoT — Coursera, UC San Diego **Oct 2024**